

Relations and Functions (10 th std)

Choose the Appropriate Answer

1. Relations and functions are subsets of the _____.
(a) Domains (b) Co – domains (c) Cartesian product (d) None of these
Ans: (c) Cartesian product
2. All constant functions are Bijections.
(a) Always true (b) Never true (c) Sometimes true (d) None of these
Ans: (b) Never true
3. Composition of functions is commutative
(a) Always true (b) Never true (c) Sometimes true (d) None of these
Ans: (c) Sometimes true
4. Composition of functions is associative
(a) Always true (b) Never true (c) Sometimes true (d) None of these
Ans: (a) Always true
5. A function can be represented by
(a) An arrow diagram (b) a set of ordered pairs
(c) a graphical form (d) All the above
Ans: (d) All the above
6. If $n(A \times B) = 15$ and $B = \{1, 3, \text{ and } 7\}$ then $n(A)$ is
(a) 3 (b) 5 (c) 1 (d) 15
Ans: (b) 5
7. If $A = \{a, b, c\}$; $B = \{b, d, e\}$; $C = \{a, e, i, o, u\}$ then $n[A \cap C] \times B$ is
(a) 18 (b) 36 (c) 9 (d) 3
Ans: (d) 3
8. If there are 28 relation from a set $A = \{2, 4, 6, 8\}$ to a set B, then the number of elements in B is
(a) 7 (b) 14 (c) 5 (d) 4
Ans: (a) 7
9. The ordered pairs $(a + 1, 4)$ $(3, 4a + b)$ are equal then (a, b) is
(a) (4, 20) (b) (20, 4) (c) (-4, 20) (d) (20, -4)
Ans: (c) (-4, 20)
10. The range of the relation $R = \{(x, x^3) / x \text{ is a prime number less than } 13\}$ is -----.
(a) (2, 3, 5, 7, 11) (b) (4, 9, 25, 49, 121)
(c) (8, 27, 125, 343, 1331) (d) (1, 8, 27, 125, 343, 1331)
Ans: (c) (8, 27, 125, 343, 1331)

11. If $\{(x, 2), (4, y)\}$ represents an identity function, then (x, y) is

- (a) (2, 4) (b) (4, 2) (c) (2, 2) (d) (4, 4)

Ans: (a) (2, 4)

12. If $\{(7, 11), (5, a)\}$ represents a constant function, then the value of 'a' is

- (a) 7 (b) 11 (c) 5 (d) 9

Ans: (b) 11

13. Given $f(x) = (-1)^x$ is a function from N to Z . Then the range of f is

- (a) $\{1\}$ (b) N (c) $\{1, -1\}$ (d) Z

Ans: (c) $\{1, -1\}$

14. If $f = \{(6, 3), (8, 9), (5, 3), (-1, 6)\}$, then the pre-images of 3 are

- (a) 5 and -1 (b) 6 and 8 (c) 8 and -1 (d) 6 and 5

Ans: (d) 6 and 5.

15. Let $A = \{1, 3, 4, 7, 11\}$, $B = \{-1, 1, 2, 5, 7, 9\}$ and $f: A \rightarrow B$ be given by

$f = \{(1, -1), (3, 2), (4, 1), (7, 5), (11, 9)\}$. Then f is

- (a) One - one (b) Onto (c) Bijective (d) Not a function

Ans: (a) One - one

16. The given diagram represents

- (a) an onto function (b) a constant function
(c) an one-one function (d) Not a function

Ans: (d) Not a function

17. If $A = \{5, 6, 7\}$, $B = \{1, 2, 3, 4, 5\}$ and $f: A \rightarrow B$ is defined by $f(x) = x - 2$, then the range of f is

- (a) $\{1, 4, 5\}$ (b) $\{1, 2, 3, 4, 5\}$ (c) $\{2, 3, 4\}$ (d) $\{3, 4, 5\}$

Ans: (d) $\{3, 4, 5\}$

18. If $f(x) = x^2 + 5$, then $f(-4) = \dots\dots\dots$

- (a) 26 (b) 21 (c) 20 (d) -20

Ans: (b) 21

19. If the range of a function is a singleton set, then it is

- (a) a constant function (b) an identity function
(c) a bijective function (d) an one-one function

Ans: (a) a constant function

20. If $f: A \rightarrow B$ is a bijective function and if $n(A) = 5$, then $n(B)$ is equal to

.....

- (a) 10 (b) 4 (c) 5 (d) 25

Ans: (c) 5

21. If $f: R \rightarrow R$ defined by $f(x) = 3x - 6$ and $g: R \rightarrow R$ defined by $g(x) = 3x + k$ if $f(x) = g(x)$. Then the value of k is

- (a) -5 (b) 5 (c) 6 (d) -6

Ans: (d) -6

22. If $f(x) = x^2 - x$ then $f(x - 1) - f(x + 1)$ is
 (a) $4x$ (b) $4x + 2$ (c) $2 - 4x$ (d) $4x - 2$

Ans: (c) $2 - 4x$

23. If $K(x) = 3x - 9$ then $L(x) = 7x - 10$ then LOK is
 (a) $21x + 73$ (b) $-21x + 73$ (c) $21x - 73$ (d) $22x - 73$

Ans: (c) $21x - 73$

$$\begin{aligned} LOK &= L[K(x)] \\ &= L(3x - 9) = 7(3x - 9) - 10 = 21x - 63 - 10 = 21x - 73 \end{aligned}$$

24. Composition of function is
 (a) Commutative (b) Associative
 (c) Commutative and Associative (d) Not associative

Ans: (b) Associative

25. A comet is heading for Jupiter with acceleration $a = 50 \text{ kms}^{-2}$. The velocity of the comet at time (t) is given by $f(t) = at^2 - at + 1$. Then the velocity at time $t = 5$ Sec is

- (a) 900 kms^{-1} (b) 1001 kms^{-1} (c) 2001 kms^{-1} (d) 50 kms^{-1}

Ans: (b) 1001 kms^{-1}

$$m = 50(5)^2 - 50(5) + 1 = 1250 - 250 + 1 = 1001 \text{ kms}^{-1}$$

26. The relation $S = \{1,1\}, \{2,2\}, \{3,3\}, \{4,4\}, \{5,5\}$ on $\{1,2,3,4,5\}$ is

- (a) Symmetric only (b) Reflexive only
 (c) Transitive only (d) An equivalence relation

Ans: (b) Reflexive only

27. Which of the following is not a type of relation?

- (a) Reflexive (b) Symmetric (c) Transitive (d) Non-linear

Ans: (d) Non-linear

28. Which of the following is an example of a reflexive relation?

- (a) $\{(1, 1), (2, 3), (3, 2)\}$ (b) $\{(1, 1), (2, 1), (1, 2)\}$
 (c) $\{(1, 2), (2, 3), (3, 1)\}$ (d) $\{(1, 2), (2, 1), (3, 3)\}$

Ans: (b) $\{(1, 1), (2, 1), (1, 2)\}$

29. Which of the following is an example of onto function?

- $f(x) = x + 1$ $f(x) = x^2$ $f(x) = |x|$ $f(x) = \sin(x)$

Ans: (a) $f(x) = x + 1$

30. Which of the following is an example of a symmetric relation?

- (a) $\{(1, 2), (2, 1), (3, 1)\}$ (b) $\{(1, 1), (2, 3), (3, 2)\}$
 (c) $\{(1, 2), (2, 3), (3, 1)\}$ (d) $\{(1, 1), (2, 2), (3, 3)\}$

Ans: (a) $\{(1, 2), (2, 1), (3, 1)\}$

31. Which of the following is an example of a one-to-one function?

- (a) $f(x) = x + 1$ (b) $f(x) = x^2$ (c) $f(x) = |x|$ (d) $f(x) = \sin(x)$

Ans: (c) $f(x) = |x|$

32. Which of the following is an example of a transitive relation?

- (a) $\{(1, 2), (2, 3), (1, 3)\}$ (b) $\{(1, 1), (2, 3), (3, 2)\}$
(c) $\{(1, 2), (2, 3), (3, 1)\}$ (d) $\{(1, 2), (2, 1), (3, 3)\}$

Ans: (a) $\{(1, 2), (2, 3), (1, 3)\}$

33. Which of the following is an example of a function that is neither onto nor one-to-one?

- (a) $f(x) = x + 1$ (b) $f(x) = x^2$ (c) $f(x) = |x|$ (d) $f(x) = \sin(x)$

Ans: (b) $f(x) = x^2$

34. Which of the following is a composite function?

- (a) $f(x) = x^2, g(x) = x + 1$ (b) $f(x) = \sin(x), g(x) = \cos(x)$
(c) $f(x) = 1/x, g(x) = x^2$ (d) $f(x) = e^x, g(x) = \ln(x)$

Ans: (a) $f(x) = x^2, g(x) = x + 1$

35. Which of the following is the inverse of the function $f(x) = 2x - 3$?

- (a) $f^{-1}(x) = x/2 - 3$ (b) $f^{-1}(x) = (x + 3) / 2$
(c) $f^{-1}(x) = 2x / 3 - 1$ (d) $f^{-1}(x) = (x - 3) / 2$

Ans: (b) $f^{-1}(x) = (x + 3) / 2$

36. Which of the following is a function that is both onto and one-to-one?

- (a) $f(x) = x^2$ (b) $f(x) = e^x$ (c) $f(x) = |x|$ (d) $f(x) = 1/x$

Ans: (b) $f(x) = e^x$

37. Which of the following is the domain of the function $f(x) = \text{Sqrt}(9 - x^2)$?

- (a) $(-3, 3)$ (b) $[-3, 3]$ (c) $(-\infty, \infty)$ (d) $[-\infty, \infty]$

Ans: (b) $[-3, 3]$

38. Which of the following is a way to test whether a function is onto?

- (a) Test whether $f(x) = 0$ for any value of x .
(b) Test whether $f(x) = f(y)$ implies $x = y$.
(c) Test whether $f(x)$ is a continuous function.
(d) Test whether every element in the range has a pre image in the domain.

Ans: (d) Test whether every element in the range has a pre image in the domain.

39. Which of the following is the inverse of the function $f(x) = \log(\text{base } 2)(x + 1)$?

- (a) $f^{-1}(x) = 2^x - 1$ (b) $f^{-1}(x) = \log(\text{base } 2)(x) - 1$
(c) $f^{-1}(x) = 2^{(x-1)} - 1$ (d) $f^{-1}(x) = \log(\text{base } 2)(x - 1)$

Ans: (a) $f^{-1}(x) = 2^x - 1$

40. Which of the following is the inverse of the function $f(x) = x^3 - 4$?

- (a) $f^{-1}(x) = (x + 4)^{(1/3)}$ (b) $f^{-1}(x) = (x + 4)^{(1/2)}$
(c) $f^{-1}(x) = (x/3 + 4)^{(1/3)}$ (d) $f^{-1}(x) = (x - 4)^{(1/3)}$

Ans: (a) $f^{-1}(x) = (x + 4)^{(1/3)}$

41. Which of the following is the domain of the function $f(x) = \ln(x^2 - 3x + 2)$?
(a) $(0, 1) \cup (2, \infty)$ (b) $(1, 2) \cup (2, \infty)$ (c) $(-\infty, 1) \cup (2, \infty)$ (d) $(-\infty, \infty)$

Ans: (c) $(-\infty, 1) \cup (2, \infty)$

42. What is the domain of the function $f(x) = \sqrt{x - 3}$?

(a) $x < 3$ (b) $x \leq 3$ (c) $x > 3$ (d) $x \geq 3$

Ans: (d) $x \geq 3$

43. What is the domain of the function $f(x) = 1 / (x + 2)$?

(a) $x \neq -2$ (b) $x < -2$ or $x > -2$ (c) $x \leq -2$ or $x \geq -2$ (d) All real numbers

Ans: (a) $x \neq -2$

44. What is the domain of the function $f(x) = \log(x - 4)$?

(a) $x < 4$ (b) $x \leq 4$ (c) $x > 4$ (d) $x \geq 4$

Ans: (c) $x > 4$

45. What is the domain of the function $f(x) = \sqrt{4 - x^2}$?

(a) $-4 < x < 4$ (b) $-4 \leq x \leq 4$ (c) $-4 \leq x < 4$ (d) $-4 < x \leq 4$

Ans: (b) $-4 \leq x \leq 4$

46. What is the domain of the function $f(x) = 1 / (x^2 - 4)$?

(a) $x \neq -2$ or $x \neq 2$ (b) $x < -2$ or $-2 < x < 2$ or $x > 2$
(c) $x \leq -2$ or $2 \leq x$ (d) All real numbers except $x = \pm 2$

Ans: (d) All real numbers except $x = \pm 2$

47. What is the domain of the function $f(x) = \sqrt{2x - 1} / (x - 4)$?

(a) $x \neq 4$ and $x > \frac{1}{2}$ (b) $x > 4$ and $x > \frac{1}{2}$
(c) $x < 4$ and $x > \frac{1}{2}$ (d) $x \neq 4$ and $x < \frac{1}{2}$

Ans: (a) $x \neq 4$ and $x > \frac{1}{2}$

48. What is the domain of the function $f(x) = \log(2x + 3) / (x^2 - 4x + 3)$?

(a) $x \neq 1$ and $x \neq 2$ (b) $x > \frac{1}{2}$ and $x > \frac{3}{2}$
(c) $x < 1$ and $x < 2$ (d) $x \neq \frac{1}{2}$ and $x \neq \frac{3}{2}$

Ans: (a) $x \neq 1$ and $x \neq 2$

49. What is the range of the function $f(x) = x^2 - 3$?

(a) All real numbers
(b) All positive real numbers
(c) All non-negative real numbers
(d) All real numbers greater than or equal to -3

Ans: (d) all real numbers greater than or equal to -3

50. What is the range of the function $f(x) = 2x + 1$?

(a) All real numbers
(b) All positive real numbers
(c) All non-negative real numbers
(d) All real numbers greater than or equal to 1

Ans: (a) All real numbers

51. What is the range of the function $f(x) = 1 / (x - 3)$?
- (a) All real numbers except 0 (b) All real numbers except 3
(c) All positive real numbers (d) All non-negative real numbers

Ans: (a) All real numbers except 0

52. Let $f(x) = x^2$ and $g(x) = 3x - 1$. What is the composite function $(f \circ g)(x)$?

(a) $9x^2 - 6x + 1$ (b) $9x^2 - 1$ (c) $3x^2 - 1$ (d) $x^2 - 3x + 1$

Ans: (a) $9x^2 - 6x + 1$

53. Let $f(x) = 2x + 1$ and $g(x) = x^2$. What is the composite function $(f \circ g)(x)$?

(a) $4x^2 + 4x + 1$ (b) $4x^2 + 2x + 1$ (c) $x^4 + 2x^2 + 1$ (d) $x^4 + 4x^2 + 1$

Ans: (a) $4x^2 + 4x + 1$

54. Let $f(x) = x - 1$ and $g(x) = x^3$. What is the composite function $(f \circ g)(x)$?

(a) $x^3 - 1$ (b) $x^3 - 2x - 1$ (c) $x^3 - x - 1$ (d) $x^4 - x - 1$

Ans: (a) $x^3 - 1$

55. The Cartesian product is also referred as

(a) Set product (b) Cross product (c) Domain product (d) None

Ans: (b) Cross product

56. In geometrical representation the product of two sets is represented in the ----.

(a) $y - z$ plane (b) $x - z$ plane (c) $x - y$ plane (d) None

Ans: (c) $x - y$ plane

57. In geometrical representation the product of three sets is represented in the ----.

(a) $y - z$ plane (b) $x - y - z$ plane (c) $x - y$ plane (d) None

Ans: (b) $x - y - z$ plane

58. Which property of sets is used for the operation of union?

(a) Associative (b) Distributive (c) Commutative (d) None

Ans: (b) Distributive

59. What are the terms used in relation functions?

(a) Domain (b) Codomain (c) Range (d) All of these

Ans: (a) Domain

60. ----- is a subset of the codomain.

(a) Range (b) Domain (c) Relation (d) None of these

Ans: (a) Range

61. Representation of relation in the form of ----.

(a) Roster (b) Set builder (c) Arrow diagram (d) All of these

Ans: (d) All of these

62. The relation is shown using connecting the elements of the sets which are contained in the box using arrows.

(a) Roster (b) Set builder (c) Arrow diagram (d) All of these

Ans: (c) Arrow diagram

63. There is no relation between any set elements is called -----.
(a) Relation (b) Function (c) Null relation (d) None of these

Ans: (c) Null relation

64. A one-to-one function, also known as an Injective function
(a) Injective function (b) Bijection (c) Onto function (d) None

Ans: (a) Injective function

65. If $A = \{1, 2\}$ and $B = \{3, 4\}$, then $A \times B$ is:
(a) $\{(1, 3), (2, 4)\}$ (b) $\{(1, 3), (1, 4), (2, 3), (2, 4)\}$
(c) $\{(3, 1), (4, 1), (3, 2), (4, 2)\}$ (d) $\{(1, 1), (2, 2)\}$

Ans: (b) $\{(1, 3), (1, 4), (2, 3), (2, 4)\}$

66. A relation R on a set A is called reflexive if:

- (a) $(a, b) \in R$ implies $(b, a) \in R$.
- (b) $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$.
- (c) $(a, a) \in R$ for all $a \in A$.
- (d) R is both symmetric and transitive.

Ans: (c) $(a, a) \in R$ for all $a \in A$.

67. A function $f: X \rightarrow Y$ is one-one (injective) if:

- (a) Every element in Y has a pre-image in X .
- (b) Different elements in X have different images in Y .
- (c) $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- (d) Both (b) and (c) are true.

Ans: (d) Both (b) and (c) are true.

68. If a function $f: X \rightarrow Y$ is both one-one and onto (surjective), it is called:

- (a) Constant function (b) Bijective function
- (c) Identity function (d) Inverse function

Ans: (b) Bijective function

69. For the function $f(x) = 2x + 1$, the domain is $\mathbb{R}$ (all real numbers). What is its range?

- (a) $\mathbb{R}$ (b) $\mathbb{R} - \{0\}$ (c) $\mathbb{R} - \{1\}$ (d) $\{1\}$

Ans: (a) $\mathbb{R}$

70. The graph of $y = 4x$ represents a line that:

- (a) Is parallel to the x -axis. (b) Is parallel to the y -axis.
- (c) Passes through the origin. (d) Is perpendicular to the y -axis.

Ans: (c) Passes through the origin

71. If a set A has 3 elements and set B has 4 elements, the total number of relations from A to B is:

- (a) $3 \times 4 = 12$ (b) $4^3 = 64$ (c) 2^{12} (d) 2^6

Ans: (c) 2^{12}

72. If $f: A \rightarrow B$ and $g: B \rightarrow C$ are functions, what is the composite function $g \circ f$?

- (a) $g(f(x))$ for all x in A (b) $f(g(x))$ for all x in B
- (c) Both (a) and (b) are correct (d) None of these

Ans: (a) $g(f(x))$ for all x in A

73. For the functions $f(x) = \sqrt{x}$ and $g(x) = x^2$, what is $(f \circ g)(x)$?

- (a) $\sqrt{x^2}$ (b) x (c) x^2 (d) $\sqrt{x^4}$

Ans: (a) $\sqrt{x^2}$

74. If $f(x) = 3x - 2$ and $g(x) = 1/x$, what is $(f \circ g)(x)$?

- (a) $3 - 2/x$ (b) $(3/x) - 2$ (c) $3/x + 2$ (d) $3 - 2x$

Ans: (b) $(3/x) - 2$

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{1}{x}\right) = 3\left(\frac{1}{x}\right) - 2 = \frac{3}{x} - 2$$

75. If $f(x) = e^x$ and $g(x) = \ln(x)$, what is $(f \circ g)(x)$?

- (a) e^x (b) x (c) $\ln(e^x)$ (d) $e^{(\ln(x))}$

Ans: (b) x

76. If $h(x) = (x^2 - 1) / (2x - 1)$, what is excluded value of x ?

- (a) -1 (b) 1 (c) 0 (d) 0.5

Ans: (d) 0.5

77. If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$, what is the symmetric difference $A \Delta B$?

- (a) $\{1, 4\}$ (b) $\{1, 2, 3, 4\}$ (c) $\{1, 2, 3\}$ (d) $\{2, 4\}$

Ans: (a) $\{1, 4\}$

78. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$ what is the complement of A with respect to B ?

- (a) $\{4, 5\}$ (b) $\{1, 2, 3\}$ (c) $\{1, 2, 4, 5\}$ (d) $\{1, 2, 3, 4, 5\}$

Ans: (a) $\{4, 5\}$

79. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, what is the Cartesian product $A \times B$?

- (a) $\{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5)\}$
(b) $\{(1,3), (2,4), (3,5)\}$
(c) $\{(1,3), (2,3), (3,3), (1,4), (2,4), (3,4), (1,5), (2,5), (3,5)\}$
(d) $\{(1,4), (2,5), (3,6)\}$

Ans: (a) $\{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5)\}$

80. What is the domain of the function $g(x) = 1 / \sqrt{x-1}$?

- (a) $x > 1$ (b) $x \geq 1$ (c) $x < 1$ (d) $x \leq 1$

Ans: (a) $x > 1$

81. What is the domain of the function $g(x) = \sqrt{x-2}$?

- (a) All real numbers (b) $x \geq 2$ (c) $x > 2$ (d) $x \geq 0$

Ans: (b) $x \geq 2$

82. In set theory, what is the Cartesian product of two sets A and B ?

- (a) The union of A and B
(b) The intersection of A and B
(c) The set of all ordered pairs (a, b) where a is in A and b is in B
(d) The complement of A with respect to B

Ans: (c) The set of all ordered pairs (a, b) where a is in A and b is in B

83. What is a relation in mathematics?

- (a) A set of ordered pairs (b) A set of natural numbers
(c) A set of real numbers (d) A set of integers

Ans: (a) A set of ordered pairs

84. What is the composition of two functions f and g , denoted as $(f \circ g)(x)$?

- (a) $f(g(x))$ (b) $g(f(x))$ (c) $f(x) + g(x)$ (d) $f(x) - g(x)$

Ans: (a) $f(g(x))$

85. If $f(x) = e^x$ and $g(x) = \ln(x)$, what is $(g \circ f)(x)$?

- (a) $e^{\ln(x)}$ (b) x (c) $\ln(e^x)$ (d) $e^{(x^2)}$

Ans: (b) x

86. If $f(x) = 2x+3$ and $g(x) = x^2-1$, what is $f \circ g(x)$?

- (a) $2x^2+5x+2$ (b) $2x^2+1$ (c) x^2+5x+2 (d) $2x^2+5x-2$

Ans: (b) $2x^2+1$

87. What is the domain of the function $f(x) = \sqrt{4-x}$?

- (a) $x \geq 2$ (b) $-2 \leq x \leq 2$ (c) $x \leq 4$ (d) $x \leq 2$

Ans: (c) $x \leq 4$

88. If $f(g(x)) = x$ for all x in the domain, what can be said about f and g ?

- (a) f is the inverse of g (b) g is the inverse of f
(c) f and g are both invertible (d) f and g are the same function

Ans: (a) f is the inverse of g

89. What is the inverse of a function?

- (a) The reflection of the graph over the x -axis
(b) A new function that "undoes" the original function
(c) The reciprocal of the original function
(d) The derivative of the original function

Ans: (b) A new function that "undoes" the original function

90. If a function is onto, what can be said about its range?

- (a) It is equal to the codomain (b) It is a subset of the codomain
(c) It is equal to the domain (d) It is a subset of the domain

Ans: (a) It is equal to the codomain

91. In a function, if each element in the domain corresponds to a unique element in the codomain, it is called:

- (a) One-to-one (b) Onto (c) Many-to-one (d) None of the above

Ans: (a) One-to-one

92. If a relation is symmetric, what can be said about (a, b) and (b, a) ?

- (a) They both belong to the relation
(b) Only (a, b) belongs to the relation
(c) Only (b, a) belongs to the relation
(d) None of the above

Ans: (a) They both belong to the relation

93. What is the cardinality of the power set of a set with n elements?

- (a) n (b) 2^n (c) $n!$ (d) n^n

Ans: (b) 2^n

94. What is the range of a function?

- (a) Set of input values (b) Set of output values
(c) Set of ordered pairs (d) Set of real numbers

Ans: (b) Set of output values

95. The domain of $\{(a,b), (b,c), (c,d)\}$ is

- (a) $\{a,b,c\}$ (b) $\{a,b\}$ (c) $\{a,b,c,d\}$ (d) $\{b,c,d\}$

Ans: (a) $\{a, b, c\}$

96. The relation $\{(a,b), (b,c), (a,d)\}$ is

- (a) Not a function (b) Function (c) Range (d) Domain

Ans: (a) Not a function

97. If $f : A \rightarrow B$ and range of $f = B$ then f is an

- (a) Objective function (b) Function (c) Onto function (d) Into function

Ans: (c) Onto function

98. The point $(-5, -7)$ lies in _____ quadrant.

- (a) I (b) II (c) III (d) IV

Ans: (c) III

99. The y coordinates of every point on the x -axis is

- (a) 1 (b) Zero (c) Negative (d) Positive

Ans: (b) Zero

100. The complement of universal set is

- (a) Impossible (b) ϕ (c) U (d) Union

Ans: (b) ϕ